

BlockVote: A decentralized voting system ensuring integrity, transparency, and trust

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Abstract

BlockVote is a secure and transparent digital voting system that leverages blockchain technology to enhance the reliability and integrity of modern elections. Traditional voting systems often face challenges such as vote tampering, lack of transparency, delayed result processing, and centralized control, which can reduce voter trust and raise concerns about fairness. BlockVote addresses these issues by implementing a decentralized blockchain network where each vote is recorded as an immutable transaction, ensuring that once a vote is cast, it cannot be altered, duplicated, or deleted. The system incorporates robust voter authentication mechanisms to ensure that only eligible users can participate, while maintaining strict voter anonymity to protect individual privacy. It enables secure vote casting through encrypted processes and provides real-time verification, allowing voters and authorities to confirm that votes have been accurately recorded. Additionally, the decentralized nature of the platform removes the need for intermediaries, reducing the risk of manipulation and increasing transparency. The system also supports efficient vote counting and instant result generation, minimizing delays and human errors. By combining security, transparency, and efficiency, BlockVote builds greater trust in the electoral process and offers a reliable solution for conducting digital elections in organizations, educational institutions, and governmental systems. Furthermore, the platform can be scaled and adapted to support large-scale national elections while maintaining consistent performance and security standards.

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Novelty:

1. Cloud-Based Zero-Knowledge Biometric Authentication

- Integrates cloud-native biometric AI services (e.g., AWS Rekognition, Azure AI Vision) for advanced voter verification.
- Captures live biometric input (face/fingerprint) from the voter through the DApp.
- Performs liveness detection to prevent spoofing attacks (e.g., photos, videos).
- Stores only a secure biometric hash on IPFS instead of raw biometric data.
- Ensures privacy preservation using zero-knowledge principles (no sensitive data exposed on-chain).
- Generates a short-lived JWT token after successful verification.
- Smart contract uses this token to authorize and unlock voting access.
- Offloads heavy computation to the cloud, improving efficiency and scalability.

2. Serverless Real-Time Analytics Dashboard

- Uses serverless cloud functions (AWS Lambda / Firebase Functions) to monitor blockchain events.
- Listens to smart contract events (e.g., `castVote`) in real time.
- Automatically triggers data processing whenever a vote is recorded.
- Stores processed data in a cloud-based NoSQL database (MongoDB Atlas).
- Enables real-time data synchronization with the frontend.
- Frontend (React / Next.js) displays live dashboards and visualizations.
- Supports analytics such as:
 - Voter turnout by region
 - Demographic insights
 - Live vote counts
- Reduces dependency on direct blockchain queries, improving speed and performance.

3. Cloud-Edge Hybrid Architecture for Low-Latency Validation

- Implements a cloud edge hybrid model for faster transaction validation.
- Deploys validation logic on edge computing platforms (Cloudflare Workers / AWS Wavelength).
- Performs a pre-flight check before sending transactions to the blockchain.
- Verifies whether a voter has already voted using a high-speed distributed cache.
- Prevents duplicate voting and spam attacks at an early stage.
- Reduces unnecessary gas fees for invalid transactions.
- Minimizes network congestion during peak voting periods.
- Ensures low latency, high throughput, and scalability for large-scale elections.

Add Ons:

- Upgradable Proxy Pattern
- IPFS + On-Chain
- 3FA (Biometric + OTP + Location)
- Gas Station Network
- React Application
- AES 512 + SHA 512